

1M1B AI for Sustainability Virtual Internship

Project Creation & Guideline Document

(In Collaboration with IBM SkillsBuild & AICTE)

1. Project Context: How to Think About Your AI + Sustainability Project

Before you begin building your project, it is important to pause and **frame your thinking correctly**.

This internship is not about building complex technology for the sake of it. It is about **learning how to observe real-world problems, think critically, and apply AI responsibly to create meaningful impact**.

As a student, you may wonder:

- What problem should I choose?
- Does my idea need to be very big or advanced?
- How do I connect AI with sustainability in a practical way?

This project is designed to help you answer those questions.

You are encouraged to start from **your surroundings**:

- Issues you notice in your campus, city, village, or community
- Problems faced by people around you
- Environmental, social, or resource-related challenges you care about

The goal is to **think like a problem-solver**, not just a technologist.

Your project should show:

- Clear thinking about the problem
- Logical use of AI tools or workflows
- Alignment with one or more Sustainable Development Goals (SDGs)
- Awareness of ethics, responsibility, and real-world impact

There is no single “right” idea. What matters most is **clarity of thought, relevance, and responsible application of AI**.

2. Purpose of This Document

This document is a step-by-step guide to help students **choose a sustainability problem, map it to an SDG, and design an AI-powered solution** as part of the 1M1B – IBM SkillsBuild AI + Sustainability Virtual Internship.

Students are encouraged to:

- Pick **any real-world problem** linked to the UN Sustainable Development Goals (SDGs)
- Apply **AI tools, models, or workflows** learned during the internship
- Build a **conceptual or working prototype** that demonstrates problem-solving using AI

There are **no restrictions on domains** as long as the problem connects to sustainability and responsible AI.

2. What Makes a Good AI + Sustainability Project

A strong project should clearly answer four questions:

1. **What problem are you solving?** (SDG-aligned)
2. **Who is affected by this problem?** (users, communities, environment)
3. **Why is AI needed?** (prediction, automation, insight, scale)
4. **How does your solution create impact?** (social, environmental, economic)

Projects may be:

- Awareness tools
- Decision-support systems
- Predictive or analytical models
- Conversational AI or assistants
- Monitoring or early-warning systems

3. Choosing Your Problem Statement (SDG Mapping)

Step 1: Select an SDG

Choose **one primary SDG** your project will address.

Examples:

- SDG 3: Good Health and Well-being
- SDG 6: Clean Water and Sanitation
- SDG 7: Affordable and Clean Energy
- SDG 11: Sustainable Cities and Communities
- SDG 12: Responsible Consumption and Production
- SDG 13: Climate Action
- SDG 15: Life on Land

You may reference **secondary SDGs**, but keep one main focus.

Step 2: Identify a Real Problem

Ask yourself:

- Is this problem visible around me (campus, city, village, industry)?
- Is data involved or generated?
- Can technology improve speed, accuracy, access, or scale?

Good Problem Statement Format:

How might we use AI to _____ so that _____ can become more sustainable?

4. Role of AI in Your Project

Your project must clearly demonstrate **where and how AI is used**.

You may use AI for:

- Classification or prediction
- Information retrieval and summarization
- Pattern detection
- Decision-making support
- Automation using agents
- Conversational interfaces

Allowed AI Components

Students may use:

- Prompt engineering
- IBM BOB
- IBM Granite Models
- Retrieval-Augmented Generation (RAG)
- Agentic AI workflows
- Entity extraction and summarization
- Multimodal inputs (text, image, document)
- Others

Projects **do not require advanced coding**. Logic, workflow design, and responsible use of AI are equally valued.

5. Project Ideation Using Design Thinking

Stage 1: Empathize

- Who faces this problem?
- What challenges do they experience?
- Why does the problem persist?

Stage 2: Define

Clearly write:

- Problem statement
- Target users
- Current gaps or inefficiencies

Stage 3: Ideate

- Brainstorm AI-based solutions
- Think of automation, prediction, or decision support
- Keep feasibility in mind

Stage 4: Prototype

- Flowchart or workflow diagram
- Prompt logic or RAG pipeline
- Sample inputs and outputs

Stage 5: Test & Refine

- Mentor feedback
- Improve clarity, ethics, and usefulness

6. Suggested Project Categories (Examples)

Environment & Climate

- Flood risk prediction assistant
- Carbon footprint estimation tool
- Climate awareness chatbot

Water & Waste

- Smart water usage advisor
- AI-based waste segregation guide
- RAG system for municipal waste policies

Energy

- Renewable energy optimization insights
- Energy consumption forecasting
- Clean energy awareness assistant

Agriculture & Biodiversity

- Crop advisory chatbot
- Pest or disease risk prediction
- Biodiversity monitoring insights

Cities & Communities

- Air quality monitoring assistant
- Sustainable transport planner
- Smart city grievance analysis tool

7. Responsible AI Guidelines (Mandatory)

Every project must address:

- **Fairness:** Avoid bias in data or assumptions
- **Transparency:** Explain how AI reaches outcomes
- **Ethics:** No harmful, discriminatory, or misleading use
- **Privacy:** Avoid personal or sensitive data misuse

Include a short section in your project:

Responsible AI Considerations

8. Final Project Deliverables

Each student must submit **PPT or PDF** with the following details:

1. Project Description

- Title
- Your Name and College Name
- SDG Alignment
- Problem Statement
- AI Solution Overview
- Target Users
- Responsible AI Considerations
- Expected Impact

2. Prototype or Demo (any one)

- Prompt workflows
- RAG demo
- Agent logic
- Screenshots or flow diagrams

3. Impact Statement (If any planning)

- What changes if this solution is implemented?
- Who benefits and how?

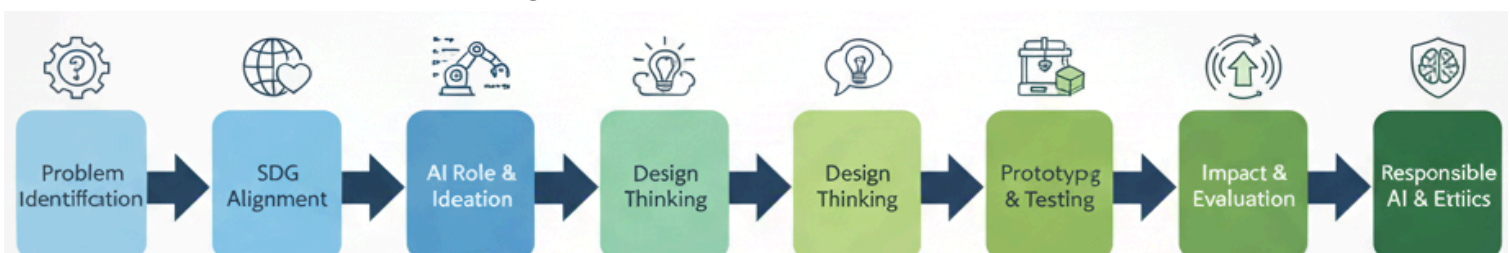
9. Key Message to Students

You are **not expected to solve the entire world's problem.**

You are expected to:

- Think critically
- Apply AI responsibly
- Design with purpose
- Build for impact

This project is your opportunity to demonstrate how **AI can be a force for sustainability, inclusion, and positive change.**



Project Making Journey: Aligned with how students think and build.